

YouTube content Analysis using SQL

Objective:

The objective of this project is to analyse YouTube video performance data using SQL. This analysis helps identify top-performing videos, audience engagement trends, category-wise performance, subscriber growth, and revenue generation.

Dataset overview:

The dataset contains YouTube video performance information including views, likes, comments, watch time, subscriber growth, upload date, category, and country.

```
select * from youtube_content;
```

	video_id integer	title character varying (100)	category character varying (50)	views integer	likes integer	comments integer	watch_time_minutes integer	subscribers_gained integer	upload_date date	country character varying (50)
1	1	SQL Tutorial	Education	50000	4500	320	120000	1500	2026-05-01	India
2	2	Python Basics	Education	70000	6200	500	180000	2300	2026-05-02	India
3	3	Gaming Live	Gaming	120000	10000	1200	250000	5000	2026-05-03	USA
4	4	Football Highlights	Sports	90000	8700	650	210000	3200	2026-05-03	UK
5	5	Movie Review	Entertainment	40000	3000	220	90000	1100	2026-05-04	India
6	6	Power BI Dashboard	Education	60000	5500	400	150000	2000	2026-05-04	Canada
7	7	Tech News	Technology	30000	2500	180	70000	900	2026-05-05	USA
8	8	Travel Vlog	Travel	80000	7600	580	190000	2800	2026-05-05	Australia

Tools used:

PostgreSQL 16

pgAdmin 4

SQL

KPIs Covered:

Total Views

Total Likes

Engagement Rate

Country-wise Analysis

Top Performing Category

Subscriber Growth

Watch Time Analysis

KPI QUERIES:

1. What is the total number of views across all YouTube videos?

```
SELECT SUM(views) AS total_views
FROM youtube_content;
```

	total_views bigint
1	540000

Helps measure overall channel reach and audience visibility.

- What is the total number of likes received across all videos?

```
SELECT SUM(likes) AS total_likes
FROM youtube_content;
```

	total_likes bigint
1	48000

Shows overall audience appreciation and engagement.

- What is the average number of views per video?

```
SELECT avg(views) AS total_views
FROM youtube_content;
```

	total_views numeric
1	67500.000000000000

Helps understand average content performance.

- Which video has highest number of views?

```
SELECT title, views
FROM youtube_content
ORDER BY views DESC
LIMIT 1;
```

	title character varying (100)	views integer
1	Gaming Live	120000

Helps identify successful content types and audience preferences.

- Which content category generates the highest views?

```
SELECT category, SUM(views) AS total_views
FROM youtube_content
GROUP BY category
ORDER BY total_views DESC;
```

	category character varying (50) 🔒	total_views bigint 🔒
1	Education	180000
2	Gaming	120000
3	Sports	90000
4	Travel	80000
5	Entertainment	40000
6	Technology	30000

Helps identify the most popular content category.

- What is the engagement rate for each video?

```
SELECT
    title,
    ROUND(((likes + comments) * 100.0 / views), 2) AS engagement_rate
FROM youtube_content;
```

	title character varying (100) 🔒	engagement_rate numeric 🔒
1	SQL Tutorial	9.64
2	Python Basics	9.57
3	Gaming Live	9.33
4	Football Highlights	10.39
5	Movie Review	8.05
6	Power BI Dashboard	9.83
7	Tech News	8.93
8	Travel Vlog	10.23

Higher engagement indicates stronger audience interaction and content quality.

- Which category gained the highest subscribers?

```
SELECT category, SUM(subscribers_gained) AS total_subscribers
FROM youtube_content
GROUP BY category
ORDER BY total_subscribers DESC;
```

	category character varying (50) 🔒	total_subscribers bigint 🔒
1	Education	5800
2	Gaming	5000
3	Sports	3200
4	Travel	2800
5	Entertainment	1100
6	Technology	900

Helps understand which content drives channel growth.

8. Which countries contribute the highest views?

```
SELECT country, SUM(views) AS total_views
FROM youtube_content
GROUP BY country
ORDER BY total_views DESC;
```

	country character varying (50) 🔒	total_views bigint 🔒
1	India	160000
2	USA	150000
3	UK	90000
4	Australia	80000
5	Canada	60000

Helps identify major audience locations for content targeting.

9. What is the average watch time across all videos?

```
SELECT AVG(watch_time_minutes) AS avg_watch_time
FROM youtube_content;
```

	avg_watch_time numeric 🔒
1	157500.00000000000000

Higher watch time indicates better audience retention.

10. How many videos were uploaded on each date?

```
SELECT upload_date, COUNT(*) AS total_videos
FROM youtube_content
GROUP BY upload_date
ORDER BY upload_date;
```

	upload_date date	total_videos bigint
1	2026-05-01	1
2	2026-05-02	1
3	2026-05-03	2
4	2026-05-04	2
5	2026-05-05	2

Helps analyse content publishing consistency.

Conclusion:

SQL can be effectively used to analyse content performance and generate business insights.